

# Production Deployment & DevOps Guide

Docker Containerization, Render Backend Deployment, Vercel SPA Hosting & Neon Cloud PostgreSQL Setup

|                   |                                                                           |
|-------------------|---------------------------------------------------------------------------|
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| Release Version:  | Version 2.0.0                                                             |
| Platform:         | PricePilot AI — AI-Powered Dynamic Pricing & Demand Forecasting SaaS      |
| Technology Stack: | Python 3.13 · FastAPI · React 19 · Scikit-Learn · PostgreSQL · SQLAlchemy |
| ML Algorithm:     | Extra Trees Regressor (Extremely Randomized Trees) — $R^2$ 0.9650         |
| Authors:          | Narendar Reddy (Lead) · Manvitha · Pravallika · Ashwindh                  |
| Organization:     | Infosys Springboard 7.0 Internship Program                                |
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## DOCUMENT REVISION HISTORY

| Rev | Date       | Author          | Reviewer           | Description                                      |
|-----|------------|-----------------|--------------------|--------------------------------------------------|
| 1.0 | 2026-08-01 | Team PricePilot | Infosys Mentor     | Initial Draft — Project Inception                |
| 1.5 | 2026-08-04 | Narendar Reddy  | Manvitha           | Mid-Sprint Review — ML Pipeline Integration      |
| 2.0 | 2026-08-07 | Full Team       | Academic Committee | Final Production Release — University Submission |

## 1.0 DEPLOYMENT ARCHITECTURE OVERVIEW

PRICEPILOT AI – PRODUCTION DEPLOYMENT TOPOLOGY Developer Machine (Windows / macOS / Linux) ■■■■ Git Push → GitHub Repository ■ ■■■■ FRONTEND: Vercel CI/CD Pipeline ■ ■■■■ npm run build → /dist (2855 modules) ■ ■■■■ Vercel Edge Network → Global CDN ■ ■■■■ URL: https://pricepilot-ai.vercel.app ■ ■■■■ BACKEND: Render Cloud Deployment ■ ■■■■ Docker Build → pricepilot-backend:2.0.0 ■ ■■■■ uvicorn main:app --host 0.0.0.0 --port 8000 ■ ■■■■ URL: https://pricepilot-api.render.com ■ ■■■■ DATABASE: Neon Cloud (Serverless PostgreSQL 16) ■■■■ Connection String: postgresql://... (env var) ■■■■ pool\_size=20, max\_overflow=10 ■■■■ Automatic SQLite fallback for local dev

Figure 1.1 — Production Deployment Topology Diagram

## 2.0 LOCAL DEVELOPMENT SETUP

### Backend Setup (Python 3.13 + FastAPI):

```
# 1. Navigate to backend directory cd PricePilot_AI/backend # 2. Create virtual environment python -m venv venv
venv\Scripts\activate # Windows source venv/bin/activate # macOS/Linux # 3. Install dependencies pip install -r requirements.txt # 4. Configure environment variables (.env)
DATABASE_URL=postgresql://user:pass@host/pricepilot SECRET_KEY=your-jwt-secret-key-minimum-32-chars
ALGORITHM=HS256 ACCESS_TOKEN_EXPIRE_MINUTES=30 # 5. Start FastAPI server uvicorn main:app --reload --port 8000
```

### Frontend Setup (Node.js 20+ + Vite 8):

```
# 1. Navigate to frontend directory cd PricePilot_AI/frontend # 2. Install Node.js dependencies npm install #
3. Configure API base URL (.env) VITE_API_BASE_URL=http://localhost:8000 # 4. Start Vite development server
npm run dev # → http://localhost:5173
```

## 3.0 DOCKER CONTAINERIZATION

```
# Dockerfile (Backend) FROM python:3.13-slim WORKDIR /app COPY requirements.txt . RUN pip install
--no-cache-dir -r requirements.txt COPY . . EXPOSE 8000 CMD ["uvicorn", "main:app", "--host", "0.0.0.0",
"--port", "8000"] # Build & Run docker build -t pricepilot-backend:2.0.0 . docker run -d -p 8000:8000
--env-file .env pricepilot-backend:2.0.0
```

## 4.0 ENVIRONMENT VARIABLES REFERENCE

| Variable Name | Required | Default                   | Description                             |
|---------------|----------|---------------------------|-----------------------------------------|
| DATABASE_URL  | YES      | sqlite:///./pricepilot.db | PostgreSQL or SQLite connection string  |
| SECRET_KEY    | YES      | —                         | JWT HS256 signing secret (min 32 chars) |

|                             |                   |                       |                                      |
|-----------------------------|-------------------|-----------------------|--------------------------------------|
| ALGORITHM                   | NO                | HS256                 | JWT algorithm (HS256 recommended)    |
| ACCESS_TOKEN_EXPIRE_MINUTES | NO                | 30                    | JWT token lifetime in minutes        |
| VITE_API_BASE_URL           | YES<br>(Frontend) | http://localhost:8000 | Backend API base URL for Axios/Fetch |

*Table 4.1 — Environment Variables Reference*